Memory and precision in noiseless Gaussian regression

Abstract

For every fixed $A > 0$, a learner that keeps at most Ad^2 bits between fresh *exact* Gaussian linear measurements needs $\Omega_A(d \log(1/\epsilon))$ measurements to estimate a uniformly random unit vector to angular error at most ϵ, for any $0 < \epsilon \leq 1/10$, with probability at least $2/3$. The constant is absolute when the memory is $o(d^2)$ bits.

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1 Introduction and main result

A single exact linear equation $y = \langle x, s\rangle$ pins the unknown vector s onto a hyperplane. With independent Gaussian rows $x_t \sim N(0, I_d)$ and exact labels $y_t = \langle x_t, s\rangle$, a handful of equations already carry arbitrary precision: if all of them could be kept, d rows would determine s almost surely. The question here is what happens to a learner that may read each pair only once, keep a bounded number of bits, and must eventually output an estimate. We show that exact observations do not rescue finite memory: sharpening ϵ forces a logarithmic-in-precision increase in the number of samples.

1.1 The model

Definition 1 (Finite-state Gaussian regression)**.** Fix integers $d \geq 2$, $M \geq 0$, $T \geq 0$. A *learner* is specified as follows. A signal $s \in S^{d-1}$ is fixed before sampling. At index $t \in \{1, \ldots, T\}$ the learner observes

$$x_t \sim N(0, I_d), \qquad y_t = \langle x_t, s\rangle,$$

where $x_1, \ldots, x_T$ are independent. The learner reads the pairs once, in order. Its persistent state always belongs to a prescribed set of at most 2^M elements. A transition may use the entire current pair (x_t, y_t) and fresh randomness, but only the next finite state is retained. The learner may stop at any index (including before the first sample) but must stop by the deterministic horizon T. At a stopping index its output $\widehat{s} \in S^{d-1}$ is generated from the terminal state, the stopping index, and fresh randomness only. All transition, stopping, and output rules may depend on d, on the target accuracy ϵ, and on the index, but never on s.

The state bound counts bits, not reals. The horizon T is fixed in advance. Accuracy means $\arccos\langle \widehat{s}, s\rangle \leq \epsilon$, i.e. the angle between the estimate and the truth is at most ϵ.

Write $\sigma = \sigma_d$ for the uniform probability measure on S^{d-1}.

1.2 Main theorem and corollaries

Theorem 2 (Memory and precision)**.** *For every fixed finite $A > 0$ there are constants $c_A > 0$ and d_A such that the following holds. Let $d \geq d_A$, $0 \leq M \leq Ad^2$ an integer, and $0 < \epsilon \leq 1/10$. Draw $S \sim \sigma$ independently of the rows and of the learner's data-independent randomness. If a learner as in Definition 1 has*

$$\mathbb{P}\{\arccos\langle \widehat{s}, S\rangle \leq \epsilon\} \geq \tfrac{2}{3},$$

then its horizon satisfies $T \geq c_A\, d \log(1/\epsilon)$.

Throughout log is natural. The constant c_A depends on the fixed number A; the statement makes no uniform claim as $A = A(d) \to \infty$.

Corollary 3 (Subquadratic memory)**.** *Let $M(d) \geq 0$ be integer-valued with $M(d) = o(d^2)$, and let $0 < \epsilon(d) \leq 1/10$ be any accuracy sequence. A family of learners with uniform-sphere average success at least $2/3$ satisfies $T(d) \geq c\, d \log(1/\epsilon(d))$ for all sufficiently large d, where $c > 0$ is absolute.*

Corollary 4 (A guarantee for every signal)**.** *The conclusions of Theorem 2 and Corollary 3 hold also when average success is replaced by the requirement that $\mathbb{P}\{\arccos\langle \widehat{s}, s\rangle \leq \epsilon\} \geq 2/3$ for* every *$s \in S^{d-1}$, the probability being over the rows and the learner's randomness.*

1.3 Why the scale $d\log(1/\epsilon)$ is natural

Before proving that memory cannot help, recall why the answer $d\log(1/\epsilon)$ is the right yardstick even *with* unlimited memory. The Kaczmarz projection step [1, 2] runs

$$z_t = z_{t-1} + \frac{y_t - \langle x_t, z_{t-1}\rangle}{\|x_t\|^2}\, x_t \qquad (x_t \neq 0),$$

and leaves $z_t = z_{t-1}$ on the Gaussian-null event $x_t = 0$. If $u_t = x_t/\|x_t\|$, the error is multiplied by $I - u_t u_t^{\mathsf{T}}$. Because the fresh direction u_t is uniform on the sphere, $\mathbb{E}(u_t u_t^{\mathsf{T}}) = I_d/d$, so conditionally

$$\mathbb{E}(\|z_t - s\|^2 \mid z_{t-1}) = (1 - 1/d)\,\|z_{t-1} - s\|^2,$$

and induction gives $\mathbb{E}\|z_t - s\|^2 = (1-1/d)^t$. Hence

$$\mathbb{E}\|z_t - s\|^2 \le e^{-t/d},$$

so $t \asymp d\log(1/\epsilon)$ samples already bring the error down to ϵ with real-valued registers. Theorem 2 says finite memory cannot beat this scale either: it does not help to be clever, and it does not help that the labels are exact.

1.4 Relation to prior work

Memory-limited learning has an established lineage. Shamir's finite-message framework [3] and the minimax analysis of Steinhardt–Duchi [4] show that retained information is itself a statistical resource. For discrete problems, Steinhardt–Valiant–Wager [5] isolated a quadratic-memory versus exponential-sample question, and Raz [6, 7] proved qualitative separations via a spectral method. For linear prediction, Dagan–Kur–Shamir [8] gave quadratic-space obstructions for a null-vector task and for a correlated random system.

The closest continuous-regression comparator is Sharan–Sidford–Valiant [9], whose Theorem 1 uses $N(0, I_d)$ covariates, a fixed unit target, and independent *uniform noise* of half-width $2^{-d/5}$. With at most $d^2/4$ bits and Euclidean accuracy d^{-r}, it yields an $\Omega(d\log r)$ lower bound over a stated range of r. Its proof draws the target uniformly on the sphere and averages over it and the samples. Our model follows that finite-width layered program, with randomized rules and bounded stopping handled directly; the decisive difference is the observation law. Noise weakens a negative result—it cannot be used to conclude anything about the *more informative* exact experiment. Theorem 2 settles the exact-observation precision question for isotropic Gaussian rows, and a short simulation (Section 10) recovers the $\Omega_A(d\,r\log d)$ consequence for the noisy "fine accuracy" range of [9].

2 The proof in one page

The proof never conditions on observed data. Conditioning the uniform prior on one exact equation almost surely confines the signal to a hyperplane section of the sphere, a law *singular* against σ; bounds would fight zero sets. We keep the original prior fixed instead and study the probability of *future* success from an arbitrary starting state.

Fix a state u at index t, reachable or not, and let $h(s)$ be the probability that, starting from (u, t) with signal s, the learner succeeds using fresh future samples. The key object is a quantitative statement about how concentrated h is. Set

$$n = d - 1, \qquad a = \frac{n}{2}, \qquad q = \left\lfloor \frac{n}{8} \right\rfloor.$$

For d large, $q \geq 2$, $q \geq d/16$, and $a \geq d/3$. We say $h : S^{d-1} \to [0,1]$ has *local mass at scale* R if

$$\int_{B(z,r)} h \, d\sigma \leq (Rr)^a \qquad \text{for all } z \in S^{d-1},\ r > 0. \tag{1}$$

The scale R is an upper-bound parameter, not a posterior radius.

The whole argument is three moves.

(i) **Terminal.** From a terminal state the output is independent of the signal. Success is confined to an ϵ-cap *globally* and to a r-cap *locally*; taking the geometric mean gives (1) at $R = \epsilon$ (Lemma 6).

(ii) **One block.** A block of $m \leq q$ rows routes to at most $N \leq (q+2)2^M$ destinations. Two facts combine: the exact joint law of (A, As) given a local restriction has a joint L^q density whose norm exceeds the input by a factor of order $\exp(Cd)R^a r^{-m(1-1/q)}$ (Theorem 10); and its *common support* is the projected ellipsoid whose average volume is at most $C_E^d r^m$ (Lemma 12). Hölder then restores the lost radius power and charges only $N^{1/q}$ for routing. Taking a-th roots, one block multiplies R by at most $\exp(Cd/a)\, N^{1/(aq)}$ (Lemma 14).

(iii) **Streaming.** Partition the stream into $B = \lceil T/q \rceil$ blocks. The per-block multiplier is at most e^Λ with

$$\Lambda = C\Big(1 + \frac{M}{d^2}\Big),$$

because $N^{1/(aq)}$ divides $\log N$ by $q \asymp d$, and $\log N$ carries the memory budget. Chaining B blocks gives the invariant at scale $R = \epsilon e^{\Lambda B}$ (Proposition 18). Since the ball of radius two covers the sphere, total success is at most $(2R)^a$.

The memory enters only through the ratio M/d^2: quadratic memory contributes a constant per block. At constant accuracy the block inequality is too weak (the logarithm is not yet large), and a separate symmetry argument supplies $T > d/4$ (Lemma 19). The two regimes join in Section 10.

3 Terminal states: how concentrated can success be?

A terminal estimate is chosen from a law independent of the signal. Its success as a function of s is concentrated on a cap, and the cap has small mass. We record both bounds and take their geometric mean.

Lemma 5 (Cap mass)**.** *For $d \geq 2$, $z \in S^{d-1}$, and $0 < r \leq 1/2$,*

$$\sigma_d(B(z,r)) \leq r^{d-1}. \tag{2}$$

Proof. Rotate z to the north pole and write $n = d-1$. For $s \in S^{d-1} \cap B(z,r)$, the first n coordinates $u = (s_1, \ldots, s_n)$ satisfy $\|u\| \leq \|s - z\| \leq r$, and the last coordinate is $s_d = \sqrt{1 - \|u\|^2} \geq \sqrt{1 - r^2} \geq 1/2$, since $r \leq 1/2$. Over the disk $\|u\| \leq r$ the graph area element $1/s_d$ is therefore at most two, so the cap has area at most $2v_n r^n$, where v_n is the volume of the unit n-ball. Each hemisphere projects onto that ball with area element $1/s_d \geq 1$, so the whole sphere has area at least $2v_n$. The ratio is at most r^n. □

Lemma 6 (Terminal local mass)**.** *Let Q be a probability law on S^{d-1} independent of the signal and put*

$$h(s) = Q\{v : \arccos\langle v, s\rangle \leq \epsilon\}.$$

Then h has local mass at scale $R = \epsilon$; that is, $\int_{B(z,r)} h \, d\sigma_d \leq (\epsilon r)^{(d-1)/2}$ for all $z \in S^{d-1}$, $r > 0$.

Proof. Fix a unit output v with $\arccos\langle v, s\rangle \le \epsilon$. Then $\|v - s\| = 2\sin(\theta/2) \le \theta \le \epsilon$, because $\sin t \le t$; so the globally successful set for a fixed v lies in a cap of chordal radius ϵ and has σ-mass at most ϵ^n. Averaging against Q gives $\int_{S^{d-1}} h\, d\sigma \le \epsilon^n$, hence $\int_{B(z,r)} h\, d\sigma \le \epsilon^n$ for every r. Now fix $z \in S^{d-1}$ and $r > 0$ (throughout the paper $\epsilon \le 1/10 < 1/2$). If $r \le \epsilon$, then $0 \le h \le 1$ and Lemma 5 give $\int_{B(z,r)} h\, d\sigma \le \sigma(B(z,r)) \le r^n \le (\epsilon r)^{n/2}$. If $r > \epsilon$, then $\int_{B(z,r)} h\, d\sigma \le \epsilon^n \le (\epsilon r)^{n/2}$. Either way the bound holds. □

What this says. Two constraints tug in opposite directions: for a small ball success cannot exceed the ball's own mass, of order r^n, and globally it cannot exceed the accuracy cap's mass ϵ^n. The geometric mean $(\epsilon r)^{n/2}$ has the same exponent in r and ϵ, exactly the form preserved by one block.

4 One block of exact equations: the projection theorem

This is the technical heart. One block of m exact rows turns a localized part of successful mass into an exact joint law of (A, As); we must control that law's density and its support. Throughout, γ_m is the law of an m-by-d standard Gaussian matrix on $\mathbb{R}^{m\times d}$, and

$$\lambda_m(dA, dy) = \gamma_m(dA)\, dy$$

is the reference measure on $\mathbb{R}^{m\times d} \times \mathbb{R}^m$. For a finite nonnegative Borel measure μ on $\mathbb{R}^d$, let η_μ be the image of $\gamma_m(dA)\,\mu(ds)$ under $(A, s) \mapsto (A, As)$. The matrix is part of the observed data; μ is not renormalized. Write v_m for the unit m-ball volume. For a closed support ball $B(z, R)$ set

$$\mathcal{E}_{z,R} = \{(A, y) : y \in Az + AB(0, R)\}. \tag{3}$$

This set is Borel (it is the zero set of a continuous function), and the exact image is carried by it.

4.1 Affine distance and the Gram determinant

Lemma 7 (Tube and inverse-distance bounds)**.** *Let $1 \le m \le d$, let $p \ge 2$ be an integer, let $\beta > m + p - 2$, $H, R > 0$, $z \in \mathbb{R}^d$, and let μ be a finite nonnegative Borel measure supported on $B(z, R)$ with $\mu(B(x, t)) \le Ht^\beta$ for all $x \in \mathbb{R}^d$, $t > 0$. If L is an affine j-plane with $0 \le j \le p - 2$, then for $0 < t \le R$*

$$\mu\{s : \operatorname{dist}(s, L) \le t\} \le 3^j 2^\beta H R^j t^{\beta-j}, \tag{4}$$

$$\int \operatorname{dist}(s, L)^{-m}\, d\mu(s) \le J_j\, HR^{\beta-m}, \qquad J_j = 1 + 3^j 2^\beta \frac{m}{\beta - j - m}. \tag{5}$$

Proof. The projection of $B(z, R)$ onto L lies in a j-dimensional ball of radius R. A maximal t-separated set in it is a t-net with at most $(1 + 2R/t)^j \le (3R/t)^j$ points: disjoint balls of radius $t/2$ fit inside the ball of radius $R + t/2$. When $j = 0$, use its single point. Any point of $B(z, R)$ within distance t of L is within $2t$ of a net point, so the ambient ball bound gives

$$\mu\{\operatorname{dist}(s, L) \le t\} \le (3R/t)^j\, H(2t)^\beta = 3^j 2^\beta H R^j t^{\beta-j},$$

which is (4). Since $\beta > j$, letting $t \downarrow 0$ also gives $\mu(L) = 0$.

The total mass is $\mu(\mathbb{R}^d) \le HR^\beta$ (take the ball $B(z,R)$). Using $\operatorname{dist}(s,L)^{-m} = R^{-m} + m\int_{\operatorname{dist}(s,L)}^{R} t^{-m-1}\,dt$ on the support (where dist $\le R$) and Tonelli,

$$\begin{aligned}\int \operatorname{dist}(s,L)^{-m}\,d\mu &\le R^{-m}\mu(\mathbb{R}^d) + m\int_0^R t^{-m-1}\mu\{\operatorname{dist}(s,L)\le t\}\,dt\\ &\le HR^{\beta-m} + 3^j 2^\beta m H R^j \int_0^R t^{\beta-j-m-1}\,dt\\ &= \Big(1 + 3^j 2^\beta \frac{m}{\beta-j-m}\Big) HR^{\beta-m} = J_j HR^{\beta-m},\end{aligned}$$

since $\beta - j - m > 0$. □

The exponent $\beta - j - m > 0$ is exactly why Theorem 10 below requires $\beta > m + p - 2$.

Lemma 8 (Gram product)**.** *Under the hypotheses of Lemma 7, for the tuple $(s_1,\dots,s_p)$ and $V = [s_2 - s_1, \dots, s_p - s_1]$,*

$$\int \det(V^\mathsf{T}V)^{-m/2}\,d\mu^{\otimes p} \le H^p R^{\beta p - m(p-1)} \prod_{j=0}^{p-2} J_j, \tag{6}$$

the integrand being assigned any fixed value on the $\mu^{\otimes p}$-null set of rank-deficient tuples.

Proof. Proper affine subspaces of dimension at most $p-2$ have μ-measure zero (by Lemma 7 in the top case $L = \mathbb{R}^d$ if $p - 1 > d$; otherwise by slicing). Hence $s_i \notin \operatorname{aff}(s_1,\dots,s_{i-1})$ for almost every tuple, so by Gram–Schmidt

$$\det(V^\mathsf{T}V)^{-1/2} = \prod_{i=2}^{p} \operatorname{dist}\big(s_i, \operatorname{aff}(s_1,\dots,s_{i-1})\big)^{-1}.$$

Integrate first in s_p, then s_{p-1}, down to s_2. At the s_i step the preceding span has dimension $i-2$ almost surely, so (2) contributes $J_{i-2}HR^{\beta-m}$. The final s_1 integral contributes at most HR^β. All integrands are nonnegative and Tonelli justifies the order, giving (6). □

4.2 The Gaussian smoothing identity

Lemma 9 (Gaussian difference identity)**.** *Let $d, m \ge 1$, $k \ge 2$, $\delta > 0$, and $t_1,\dots,t_k \in \mathbb{R}^d$. Put $D = [t_2 - t_1, \dots, t_k - t_1]$ and let ϕ_δ be the $N(0,\delta^2 I_m)$ density. If $A \sim \gamma_m$ then*

$$\mathbb{E}_A \int_{\mathbb{R}^m} \prod_{i=1}^{k} \phi_\delta(y - At_i)\,dy = (2\pi)^{-m(k-1)/2} \det\Big(D^\mathsf{T}D + \delta^2(I_{k-1} + \mathbf{1}\mathbf{1}^\mathsf{T})\Big)^{-m/2}. \tag{7}$$

If D has full column rank, the right side is at most $(2\pi)^{-m(k-1)/2}\det(D^\mathsf{T}D)^{-m/2}$.

Proof. Introduce independent $N(0,\delta^2 I_m)$ vectors η_i. For fixed A, integrating $\prod_i \phi_\delta(y - At_i)$ over the common y gives the density at zero of the $k-1$ differences $(At_i + \eta_i) - (At_1 + \eta_1)$, $2 \le i \le k$; the change of variables from k vectors to the first plus these differences has determinant one. Averaging over A, each of the m rows of these differences is a centered Gaussian vector with covariance $D^\mathsf{T}D + \delta^2(I + \mathbf{1}\mathbf{1}^\mathsf{T})$, and the rows are independent; evaluating the joint density at zero gives (7). If $G = D^\mathsf{T}D$ is positive definite and Q positive semidefinite, $\det(G+Q) = \det G\,\det(I + G^{-1/2}QG^{-1/2}) \ge \det G$. Apply with $Q = \delta^2(I + \mathbf{1}\mathbf{1}^\mathsf{T})$. □

4.3 The exact projection density

Theorem 10 (Exact projection density)**.** *Let $d, m \geq 1$ and $p \geq 2$ be integers, let $\beta > m + p - 2$, $H, R > 0$, $z \in \mathbb{R}^d$, and let μ be a finite nonnegative Borel measure supported on $B(z, R)$ with $\mu(B(x,t)) \leq Ht^\beta$ for all x, t. Then η_μ has a nonnegative jointly Borel density F with respect to λ_m, which can be taken to vanish outside $\mathcal{E}_{z,R}$, and*

$$\|F\|_{L^p(\lambda_m)} \leq \mathcal{B} := (2\pi)^{-m(p-1)/(2p)}\, H R^{\beta - m(1-1/p)} \Big(\prod_{j=0}^{p-2} J_j\Big)^{1/p}, \tag{8}$$

with J_j as in Lemma 7.

Proof. **Step 1 (smoothing moment).** For $\delta > 0$ put $F_\delta(A, y) = \int \phi_\delta(y - As)\, d\mu(s)$. Expanding the p-th power and using Tonelli and Lemma 9 with $k = p$,

$$\int F_\delta^p \, d\lambda_m = (2\pi)^{-m(p-1)/2} \int \det\Big(V^{\mathsf{T}} V + \delta^2 (I + \mathbf{1}\mathbf{1}^{\mathsf{T}})\Big)^{-m/2} \, d\mu^{\otimes p} \leq \mathcal{B}^p, \tag{9}$$

where we used that adding the positive semidefinite covariance only increases the determinant, that rank-deficient tuples are null for $\mu^{\otimes p}$, and Lemma 8. Note the exact $(2\pi)^{-m(p-1)/2}$ under the root matches the prefactor in (8).

Step 2 (limit functional). For real $\psi \in C_c(\mathbb{R}^{md} \times \mathbb{R}^m)$ and independent $Z \sim N(0, I_m)$,

$$\int \psi F_\delta \, d\lambda_m = \int \mathbb{E}_Z \psi(A, As + \delta Z)\, d\gamma_m(A)\, d\mu(s) \longrightarrow \int \psi \, d\eta_\mu$$

by bounded convergence, since μ is finite and ψ is bounded continuous. By Hölder and (9), $\big|\int \psi F_\delta \, d\lambda_m\big| \leq \mathcal{B}\, \|\psi\|_{L^{p/(p-1)}(\lambda_m)}$.

Step 3 (density by duality). The measure λ_m is locally finite, σ-finite, Borel; η_μ is finite, nonnegative, Borel. The bounded functional $\psi \mapsto \int \psi \, d\eta_\mu$ on C_c extends to $L^{p/(p-1)}(\lambda_m)$ with norm at most $\mathcal{B}$. Standard L^p duality then gives $f \in L^p(\lambda_m)$ with $\|f\|_p \leq \mathcal{B}$ and $\eta_\mu = f\lambda_m$; taking a Borel representative and using $\eta_\mu = f^+ \lambda_m$ (the negative part is a nonnegative measure that annihilates all nonnegative tests) yields a nonnegative density.

Step 4 (support). The image measure is carried by the Borel set $\mathcal{E}_{z,R}$; multiplying the density by $\mathbf{1}_{\mathcal{E}_{z,R}}$ preserves the identity and makes F vanish almost everywhere outside it. □

Remark 11. Theorem 10 is the only place measure-theoretic care is needed. The analytic content is entirely in Lemma 7 and Lemma 8; the passage from the smoothed bound (9) to the exact density is a standard Riesz-representation argument, and we have included it so that the proof uses *exact* labels, not a smoothed surrogate.

5 How much room do the labels take?

The density bound of Theorem 10 loses $m(1 - 1/p)$ powers of the support radius; the same powers must be recovered by the average volume of the common support. The projected ellipsoid does exactly this.

Lemma 12 (Projected volume)**.** *Let $1 \leq m \leq d$, $z \in \mathbb{R}^d$, $R > 0$, and let $\mathcal{E}_{z,R}$ be as in (3). There is an absolute $C_E \geq 1$ with, for a standard Gaussian m-by-d matrix,*

$$\lambda_m(\mathcal{E}_{z,R}) = v_m R^m\, \mathbb{E}_A \sqrt{\det(AA^{\mathsf{T}})} \leq v_m R^m d^{m/2} \leq R^m \Big(\frac{2\pi e\, d}{m}\Big)^{m/2} \leq C_E^d R^m. \tag{10}$$

Proof. Almost surely AA^{T} is invertible; in singular-value coordinates $AB(0,R)$ is an ellipsoid with semiaxes R times the singular values of A, so its volume is exactly $v_m R^m \sqrt{\det(AA^{\mathsf{T}})}$. Hadamard's inequality bounds $\sqrt{\det(AA^{\mathsf{T}})}$ by the product of the row norms, whose expectations are at most $\sqrt{d}$ each; independence gives the first inequality. Also $(2\pi)^{m/2} = \int e^{-\|y\|^2/2}\,dy \geq e^{-m/2} v_m m^{m/2}$, whence $v_m \leq (2\pi e/m)^{m/2}$. For $t = m/d \in (0,1]$, since $t\log(1/t) \leq 1/e$,

$$\frac{1}{d}\log\Big(\big(\tfrac{2\pi e d}{m}\big)^{m/2}\Big) = \frac{t}{2}\log(2\pi e) + \frac{t}{2}\log\tfrac{1}{t} \leq \frac{1}{2}\log(2\pi e) + \frac{1}{2e},$$

an absolute constant, so some C_E works. □

6 Routing among many destinations

All later estimates select among boundedly many continuation measures supported on one common data domain. Selection costs only $N^{1/p}$.

Lemma 13 (Selection)**.** *Let $(E,\mathcal{E},\lambda)$ be a measure space, $\mathcal{D}\in\mathcal{E}$ of finite measure, $p>1$, $U\geq 0$, $N\geq 1$. Suppose finite nonnegative measures $\eta_1,\dots,\eta_N$ are supported on $\mathcal{D}$ and satisfy, for all measurable $0\leq g\leq 1$,*

$$\int g\,d\eta_j \leq U\Big(\int_{\mathcal{D}} g\,d\lambda\Big)^{1-1/p}. \tag{11}$$

Then for measurable $g_j\geq 0$ with $\sum_j g_j \leq 1$,

$$\sum_{j=1}^{N}\int g_j\,d\eta_j \leq N^{1/p} U\,\lambda(\mathcal{D})^{1-1/p}. \tag{12}$$

In particular, if $U\leq HLr^{\beta-m(1-1/p)}$ and $\lambda(\mathcal{D})\leq Vr^m$, then the right side is at most $N^{1/p}HV^{1-1/p}L\,r^{\beta}$.

Proof. Let $b_j = \int_{\mathcal{D}} g_j\,d\lambda$. Each $g_j\in[0,1]$, so (11) gives $\int g_j\,d\eta_j \leq U b_j^{1-1/p}$, and Hölder on the finite index set gives $\sum_j U b_j^{1-1/p} \leq U N^{1/p}(\sum_j b_j)^{1-1/p}$. Since $\sum_j g_j \leq 1$, we have $\sum_j b_j \leq \lambda(\mathcal{D})$; this proves (12). The radius substitution multiplies the two powers of r into r^{β}. □

7 Putting one block together

Lemma 14 (Backward block bound)**.** *Let $1\leq m\leq q$ and $N\geq 1$. Suppose $h_1,\dots,h_N : S^{d-1}\to[0,1]$ have local mass at a common scale $R>0$ in the sense of (1). Let $g_j:\mathbb{R}^{m\times d}\times\mathbb{R}^m\to[0,1]$ be Borel with $\sum_j g_j\leq 1$ and define $h(s)=\sum_j h_j(s)\int g_j(A,As)\,d\gamma_m(A)$. Then for every $z\in S^{d-1}$ and $r>0$,*

$$\int_{B(z,r)} h\,d\sigma \leq \exp(Cd)N^{1/q}(Rr)^a. \tag{13}$$

Equivalently h has local mass at scale $R' = R\exp(Cd/a)N^{1/(aq)}$.

Proof. Fix $z\in S^{d-1}$, $r>0$, and $\mu_j = \mathbf{1}_{B(z,r)}h_j\sigma$, a finite Borel measure on $B(z,r)$. If an ambient ball $B(x,t)$ has positive μ_j-mass, choose a point p of its intersection with the sphere; every other point of the intersection lies within $2t$ of p. The local bound therefore gives

$$\mu_j(B(x,t)) \leq (2Rt)^a \qquad (x\in\mathbb{R}^d,\ t>0). \tag{14}$$

Apply Theorem 10 with exponent $\beta = a$, moment order $p = q$, coefficient $H = (2R)^a$, and support radius r. The margin $J_\ell \le \exp(Cd)$ holds for $0 \le \ell \le q-2$ by (1) parameters; absorbing 2^a, we obtain exact joint densities f_j, supported on $\mathcal{E}_{z,r}$, with

$$\|f_j\|_{L^q(\lambda_m)} \le \exp(Cd) R^a r^{a-m(1-1/q)}.$$

These represent unnormalized successful mass, not posteriors.

Put $q' = q/(q-1)$. By the exact image-measure identity, Hölder on matrix–label space, and Hölder on the finite index set,

$$\begin{aligned}
\int_{B(z,r)} h \, d\sigma &= \sum_j \int_{\mathcal{E}_{z,r}} f_j g_j \, d\lambda_m \\
&\le \exp(Cd) R^a r^{a-m(1-1/q)} \sum_j \Big(\int_{\mathcal{E}_{z,r}} g_j^{q'} \, d\lambda_m \Big)^{1/q'} \\
&\le \exp(Cd) R^a r^{a-m(1-1/q)} N^{1/q} \Big(\int_{\mathcal{E}_{z,r}} \sum_j g_j^{q'} \, d\lambda_m \Big)^{1/q'}.
\end{aligned}$$

Since $0 \le g_j \le 1$ and $\sum_j g_j \le 1$, the last integral is at most $\lambda_m(\mathcal{E}_{z,r}) \le C_E^d r^m$ by Lemma 12. Its $1 - 1/q$ power restores the radius exponent: $r^{a-m(1-1/q)}(r^m)^{1-1/q} = r^a$. Absorbing constants proves (13); the a-th root gives the scale update. □

Remark 15 (Why the two radii cancel). The localization radius r appears with *opposite* signs in the density norm ($r^{-m(1-1/q)}$) and the support volume ($r^{+m(1-1/q)}$). Their product leaves exactly r^a, so the block bound pays no price for localizing. Only the destination count and absolute constants remain. For a block of length $m \le q$ with $N \le (q+2)2^M$,

$$\log \frac{R'}{R} \le \frac{Cd}{a} + \frac{M \log 2 + \log(q+2)}{aq} \le C\Big(1 + \frac{M}{d^2}\Big) =: \Lambda,$$

since $d/a \le 3$ and $1/(aq) \le 48/d^2$. Here is where quadratic memory enters: $q \asymp d$ divides the log of the number of choices.

8 Reading a whole stream

We now split the horizon into blocks of at most q rows and iterate Lemma 14. Two bookkeeping facts are needed: how many destinations a block can have, and how to pass from completed-measurable rules to Borel rules without changing the prior experiment.

8.1 Destinations of a block

Lemma 16 (Destinations)**.** *For a fixed-seed Borel learner with state sets Q_t ($|Q_t| \le 2^M$), a sample-free stopping check at each index, and the horizon treated as terminal, let $0 \le t < T$, $u \in Q_t$, and $1 \le m \le T - t$. There are an integer $1 \le N \le (m+2)2^M$, Borel routes $g_j \ge 0$ with $\sum_j g_j \le 1$, and Borel functions $h_j : S^{d-1} \to [0,1]$ such that, for every s,*

$$h_{t,u}(s) = \sum_{j=1}^{N} h_j(s) \int g_j(A, As) \, d\gamma_m(A). \tag{15}$$

Each destination is either a terminal state (with output law depending only on its state and stopping index) or a continuing state at index $t + m$.

Proof. Pre-generate all m rows of the block. At each $k = t, \dots, t+m-1$ perform the sample-free stopping check; if active, process the next pair into the next state. A terminal tag from the last transition stops at $t+m$; an active exit is passed to the continuation before the check at $t+m$, so that check is performed exactly once. A terminal destination records its local offset $i \in \{0, \dots, m\}$ and state in Q_{t+i} (with $i = 0$ the immediate stop and $i = m$ from the last transition); these outcomes merge with same-state, same-index sample-free stops. Continuing destinations form one extra copy of Q_{t+m}, distinguished from terminal ones at that index. Thus

$$N \le \sum_{i=0}^{m} |Q_{t+i}| + |Q_{t+m}| \le (m+2)2^M.$$

Because only local offsets appear, the count has no factor of T. For arbitrary data (A, b), each $g_j(A, b)$ is a finite sum of products of Borel coordinate probabilities, hence Borel; conditioning on block data then on the destination proves (15) for every signal and every starting state. □

8.2 Borel versions

Lemma 17 (Borel reduction)**.** *Let $d \ge 2$, let π be a Borel probability measure on S^{d-1}, and let $X \sim \gamma_1$ be independent of $S \sim \pi$. Suppose*

$$\mathcal{L}(X, \langle X, S\rangle) \ll \lambda, \qquad \lambda(dx, dy) = \gamma_1(dx)\, dy. \tag{16}$$

Fix a horizon $T \ge 0$ and finite state sets. If every coordinate of the finite transition/stopping vector is measurable in the λ-completion of the Borel σ-field, then there are everywhere-defined Borel transition and stopping kernels, with the same state sets and terminal output laws, whose π-prior joint law of the signal, state path, stopping index and output is unchanged. The domination (16) holds when $\pi = \sigma$ is uniform, for every $d \ge 2$.

Proof. If f is λ-completed-measurable, each dyadic superlevel set agrees with a Borel set off a Borel λ-null set; replacing these by their Borel representatives in $f_n = 2^{-n} \sum_{k=1}^{2^n} \mathbf{1}\{f \ge k2^{-n}\}$ yields Borel f_n with a Borel limit equal to f off the union of countably many null sets. Do this simultaneously for all coordinates of all kernels; the (null) set where a resulting vector fails to be a probability vector is repaired by a fixed point mass. One Borel λ-null set Z contains every altered pair. Pre-generate all T rows; each pair $(X_t, \langle X_t, S\rangle)$ has λ-null Z-preimage, and a finite union over t shows no altered pair is ever visited almost surely. Coupling original and modified finite choices by shared uniform randomness makes the state paths agree almost surely; equal terminal state and index give equal output laws.

For the domination: condition on $X = x$, rotate to make $x/\|x\|$ the first axis, and write a uniform sphere point as $G/\|G\|$ with $G = (G_1, G_\perp)$. Then $\|G_\perp\| > 0$ almost surely and is independent of G_1, and $g \mapsto g/\sqrt{g^2 + R^2}$ ($R = \|G_\perp\|$) is a smooth increasing bijection of $\mathbb{R}$ with nonzero derivative. So $\langle X, S\rangle$ is absolutely continuous for every $x \ne 0$; Fubini over sections of a λ-null set proves (16). □

The reduction averages original conditional success probabilities; it never exchanges limits with a jointly measurable choice of Borel replacements.

8.3 The quantitative success bound

Proposition 18 (Success after a finite stream)**.** *There are absolute constants $C, d_0 > 0$ with the following property. For $d \ge d_0$, every learner with $M \ge 0$ bits and integer horizon $T \ge 0$ has, for $S \sim \sigma$ independent of the fresh Gaussian rows,*

$$\mathbb{P}\{\arccos\langle \widehat{s}, S\rangle \le \epsilon\} \le \min\Big\{1,\ \big[2\epsilon \exp\big(\Lambda\lceil T/q\rceil\big)\big]^a\Big\}, \tag{17}$$

where $0 < \epsilon \le 1/10$, $a = (d-1)/2$, $q = \lfloor (d-1)/8 \rfloor$, *and* $\Lambda = C(1 + M/d^2)$.

Proof. Use the angular score. Fix a prior-preserving Borel replacement (Lemma 17) with prior σ. If $T = 0$, the output is a signal-independent mixture of terminal laws, so total success is at most $(2\epsilon)^a$. Now assume $T \ge 1$; increase d_0 so $q \ge 2$, $q \ge d/16$, $a \ge d/3$. Partition the T indices into $B = \lceil T/q \rceil$ consecutive blocks of length at most q. At the final boundary every continuation is terminal and obeys Lemma 6 at scale ϵ.

Suppose at the end of one block all continuing functions obey (1) at a common scale $R \ge \epsilon$; terminal destinations do so too by Lemma 6 and monotonicity in the scale. For a block of length $m \le q$, Lemma 16 puts every preceding continuation in the form required by Lemma 14 with $N \le (q+2)2^M$. That lemma multiplies the scale by at most $\exp(C_0 d/a) N^{1/(aq)}$, whose logarithm is at most

$$\frac{C_0 d}{a} + \frac{M \log 2 + \log(q+2)}{aq} \le C\Big(1 + \frac{M}{d^2}\Big) = \Lambda,$$

using $d/a \le 3$, $1/(aq) \le 48/d^2$, and $\log(q+2)/(aq) \le 48 \log(d+2)/d^2$. Backward induction gives (1) at scale $R = \epsilon e^{\Lambda B}$ for every initial state; averaging over the independent initial-state law preserves it. The radius-two ball about any sphere point covers S^{d-1}, so total success is at most $(2R)^a$. By Lemma 17 this bounds the original conditional success for almost every seed; averaging over seeds gives (17), and a probability is also at most one. □

9 Constant accuracy: a symmetry argument

For ϵ not small, the block inequality is too lossy; but a learner that sees only T rows can never distinguish a signal from its reflection across the row span. Even with unlimited memory and all data, accuracy near $1/10$ needs more than $d/4$ rows.

Lemma 19 (Reflection)**.** *Let $d \ge 1$, $T \ge 0$, $S \sim \sigma$. Let a T-by-d matrix A, independent of S, have row span U with $\dim U \le T$, and let $\widehat{S}$ be drawn from a kernel depending only on (A, AS). Put $u = P_U S$, $v = (I - P_U)S$. For any $\widehat{S} \in \mathbb{R}^d$ and $0 < \epsilon < 1$,*

$$\mathbb{P}\{\|\widehat{S} - S\| \le \epsilon\} \le \frac{1}{2} + \frac{T}{2d(1-\epsilon^2)}. \tag{18}$$

For a unit output, $\mathbb{E}\langle \widehat{S}, v\rangle = 0$ and $\mathbb{E}\langle \widehat{S}, S\rangle \le \sqrt{T/d}$.

Proof. Conditional on A, reflection across U sends $S = u + v$ to $u - v$. It preserves uniform spherical measure and the data (A, AS), because $Av = 0$, so the output kernel is the same at both signals. If $\|v\| > \epsilon$, the two reflected signals are more than 2ϵ apart; a fixed estimate succeeds for at most one, so averaging the common kernel over the pair gives success at most $1/2$ on that event. Hence

$$\mathbb{P}\{\|\widehat{S} - S\| \le \epsilon\} \le \frac{1}{2} + \frac{1}{2}\mathbb{P}\{\|u\|^2 \ge 1 - \epsilon^2\}.$$

Spherical isotropy and independence of A give $\mathbb{E}(\|u\|^2 \mid A) = \dim U/d \le T/d$; Markov proves (18). For unit outputs reflection cancels $\mathbb{E}\langle \widehat{S}, v\rangle$, while $\langle \widehat{S}, u\rangle \le \|u\|$; Cauchy–Schwarz and the same second moment give the expectation bound. □

Corollary 20. *If $T \le d/4$ and $\epsilon \le 1/10$, then any learner (even with unlimited memory and all pre-generated rows) has chordal, hence angular, success probability less than $2/3$.*

Proof. By (18), success is at most $\frac{1}{2} + \frac{1}{8(1-\epsilon^2)} \le \frac{1}{2} + \frac{1}{7.92} < \frac{2}{3}$. Angular error $\le \epsilon$ implies $\|\widehat{S} - S\| \le \epsilon$ since $2\sin(\theta/2) \le \theta \le \epsilon$. A learner with a deterministic horizon uses a kernel of the full data, so the bound applies. □

10 Proof of the main theorem

Proof of Theorem 2. Fix $A > 0$. Let d exceed the absolute threshold of Proposition 18 and satisfy $a \geq d/3$, $q \geq d/16$. Put

$$K_A = \exp(C(1+A)) > 1, \qquad L = \log(1/\epsilon), \qquad B = \lceil T/q \rceil,$$

with C the absolute constant in Λ. Since $M \leq Ad^2$ we have $\Lambda \leq C(1+A) = \log K_A$. If prior success is at least $2/3$, then by (17)

$$\tfrac{2}{3} \leq (2\epsilon K_A^B)^a,$$

so $\epsilon K_A^B \geq (1/2)(2/3)^{1/a} \geq 1/3$ for large d (since $a \geq d/3$). Taking logarithms and using $B \leq T/q+1$,

$$T \geq q\Big(\frac{L - \log 3}{\log K_A} - 1\Big). \tag{19}$$

Let $H_A = 2(\log 3 + \log K_A)$. If $L \geq H_A$, then $T \geq qL/(2\log K_A) \geq dL/(32 \log K_A)$. If $L < H_A$, Corollary 20 gives $T > d/4 > dL/(4H_A)$. Thus with

$$c_A = \min\Big\{\frac{1}{32\log K_A},\ \frac{1}{4H_A}\Big\} > 0$$

we have $T \geq c_A\, d \log(1/\epsilon)$ throughout $0 < \epsilon \leq 1/10$, $M \leq Ad^2$. □

Proof of Corollary 3. Since $M(d) = o(d^2)$, there is d_M with $M(d) \leq d^2$ for $d \geq d_M$. Apply Theorem 2 with $A = 1$ for each $d \geq d_M$ and each accuracy $0 < \epsilon(d) \leq 1/10$; $K_1 = e^{2C}$ is absolute and $c_1 > 0$ is absolute. The dimension threshold may depend on the memory sequence through d_M. No relation between memory and accuracy is needed. □

Proof of Corollary 4. Average the fixed-signal experiment over $S \sim \sigma$ before selecting Borel versions; measurability makes the averaged success the integral of the per-signal successes, so success $\geq 2/3$ for every signal gives uniform success $\geq 2/3$. Theorem 2 and Corollary 3 then apply with unchanged constants and thresholds; Lemma 17 preserves the average even if a version changes behavior at exceptional individual signals. □

Remark 21 (The noisy-experiment consequence). Theorem 2 yields an $\Omega_A(d\, r \log d)$ lower bound in the noisy Gaussian experiment of [9] for fixed A, memory $M \leq Ad^2$, and Euclidean accuracy $e = d^{-r}$ in the stated fine-accuracy range. An exact-observation learner generates the prescribed independent uniform noise inside each transition and simulates the noisy learner with the same states and horizon (the per-seed Borel reduction applies first, since the noisy pair law has a density). On $\|V - S\| \leq e \leq 1/20$, normalizing V gives angular error at most $\arcsin e \leq 2e$; taking $\epsilon = 2e$ gives $T \geq c_A d \log(1/(2e)) = \Omega_A(d\, r \log d)$.

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